EXPERIMENT 5 (SURAJ NARASIMHAN POTTY_192412442)

1. Aim

To analyze the S-parameters of a microwave directional coupler and a two-way power divider at 10 GHz and determine coupling factor, isolation, insertion loss, directivity, and power-division ratio using MATLAB.

Objective

To simulate and analyze a microwave directional coupler and two-way power divider at 10 GHz using MATLAB, and determine their coupling, isolation, insertion loss, directivity, and power-division ratio.

2. Apparatus Required

1. Computer or laptop
2. MATLAB software

3. Procedure

1. Open MATLAB and create a new script.
2. Set the design frequency to 10 GHz and reference impedance to 50 Ω.
3. Define the S-parameters of the directional coupler and two-way power divider.
4. Calculate coupling factor, isolation, insertion loss, directivity, and transmission efficiency.
5. Plot the S-parameter responses over the 8–12 GHz frequency range.
6. Change the coupling-section length by 5% and compare the responses before and after the change.
7. Observe the graphs and identify the design performance at 10 GHz.

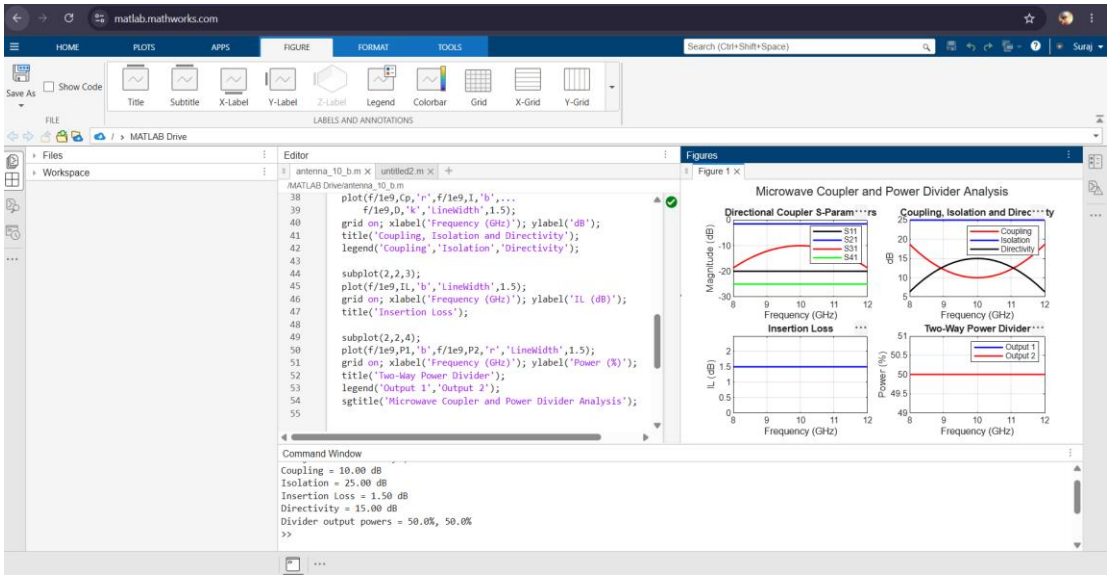
4. MATLAB Program

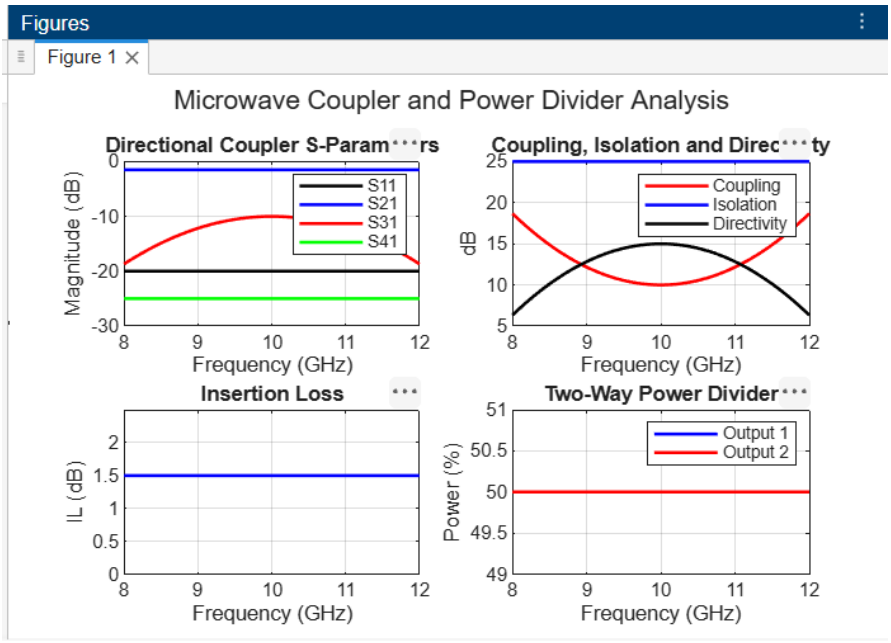
This program provides a simplified idealized model of the coupler and divider, with illustrative frequency-dependent responses. It is not a full electromagnetic simulation in CST or HFSS.

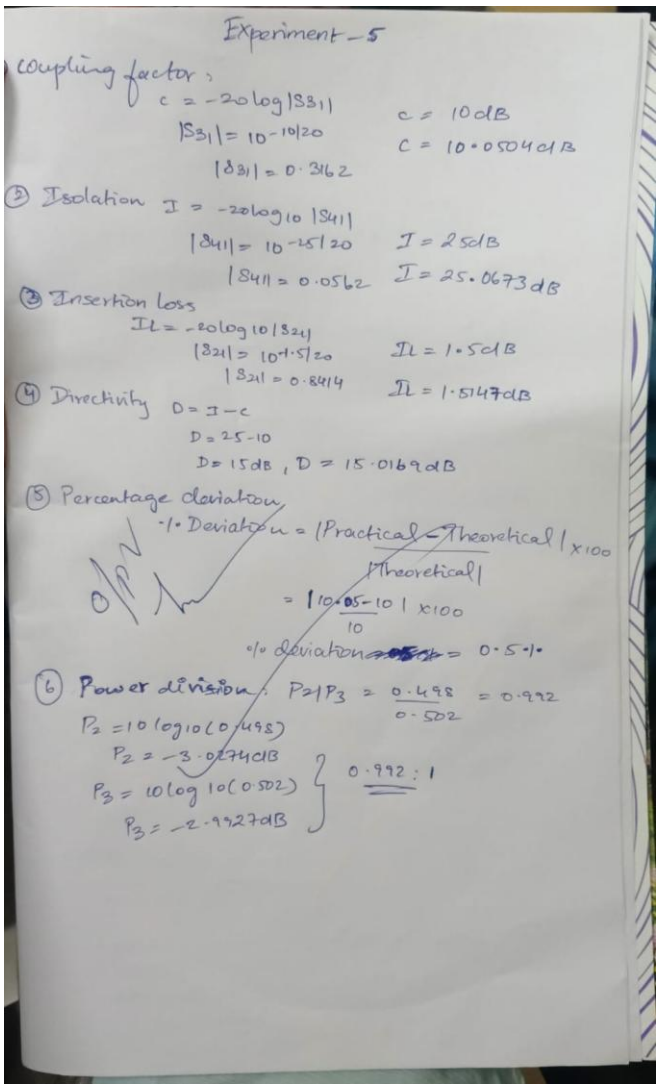
```
clc; clear; close all;
f=linspace(8e9,12e9,201); f0=10e9;
C=10; Z0=50;
S21=10^(-1.5/20)*ones(size(f));
S31=10^(-C/20)*exp(-((f-f0)/2e9).^2);
S41=10^(-25/20)*ones(size(f));
S11=10^(-20/20)*ones(size(f));
I=-20*log10(abs(S41));
Cp=-20*log10(abs(S31));
IL=-20*log10(abs(S21));
D=I-Cp;
S21d=ones(size(f))/sqrt(2);
S31d=ones(size(f))/sqrt(2);
P1=abs(S21d).^2*100;
P2=abs(S31d).^2*100;
fprintf('Suraj Narasimhan Potty | 192412442\n');
fprintf('Coupling = %.2f dB\n',Cp(101));
fprintf('Isolation = %.2f dB\n',I(101));
fprintf('Insertion Loss = %.2f dB\n',IL(101));
fprintf('Directivity = %.2f dB\n',D(101));
fprintf('Divider output powers = %.1f%%, %.1f%%\n',...
P1(101),P2(101));
figure('Color','w','Position',[100 100 1000 650]);
subplot(2,2,1);
plot(f/1e9,20*log10(abs(S11)),'k',...
f/1e9,20*log10(abs(S21)),'b',...
f/1e9,20*log10(abs(S31)),'r',...
f/1e9,20*log10(abs(S41)),'g','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');
```

```
title('Directional Coupler S-Parameters');
legend('S11','S21','S31','S41');
subplot(2,2,2);
plot(f/1e9,Cp,'r',f/1e9,I,'b',...
f/1e9,D,'k','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('dB');
title('Coupling, Isolation and Directivity');
legend('Coupling','Isolation','Directivity');
subplot(2,2,3);
plot(f/1e9,IL,'b','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('IL (dB)');
title('Insertion Loss');
subplot(2,2,4);
plot(f/1e9,P1,'b',f/1e9,P2,'r','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Power (%)');
title('Two-Way Power Divider');
legend('Output 1','Output 2');
sgtitle('Microwave Coupler and Power Divider Analysis');
```

OUTPUT:







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Coupling factor:

$C = -20\log|S_{31}|$

$|S_{31}| = 10^{-10/20}$

$|S_{31}| = 0.3162$

$C = 10\,dB$

$C = 10.0504\,dB$

(2) Isolation $I = -20\log_{10}|S_{41}|$

$|S_{41}| = 10^{-25/20}$

$|S_{41}| = 0.0562$

$I = 25\,dB$

$I = 25.0673\,dB$

(3) Insertion loss

$IL = -20\log_{10}|S_{21}|$

$|S_{21}| = 10^{-1.5/20}$

$|S_{21}| = 0.8414$

$IL = 1.5\,dB$

$IL = 1.5147\,dB$

(4) Directivity $D = I - C$

$D = 25 - 10$

$D = 15\,dB$, $D = 15.0169\,dB$

(5) Percentage deviation

% Deviation = $\left|\frac{\text{Practical} - \text{Theoretical}}{\text{Theoretical}}\right| \times 100$

$= \left|\frac{10.05 - 10}{10}\right| \times 100$

% deviation = 0.5%

(6) Power division: $P_2/P_3 = \frac{0.498}{0.502} = 0.992$

$P_2 = 10\log_{10}(0.498)$

$P_2 = -3.0274\,dB$

$P_3 = 10\log_{10}(0.502)$

$P_3 = -2.9927\,dB$

$0.992 : 1$

5. Reflection

I learned how to analyze the S-parameters of microwave directional couplers and power dividers. I understood how coupling, isolation, insertion loss, directivity, and power-division ratio are used to evaluate microwave component performance.

6. Result

The S-parameter characteristics of a microwave directional coupler and a two-way power divider were analyzed using MATLAB. The coupling factor, isolation, insertion loss, directivity, and output power division were calculated, and their frequency responses were plotted.